

Power BI Ungraded Project – Data Cleaning & Transformation Using Power Query

Assignment Tasks

1. Load Data into Power BI

Steps:

- Open Power BI and **load the dataset** from a CSV file.
- Open **Power Query Editor** to begin cleaning the data.

Expected Output:

- The dataset is successfully loaded into Power BI's Power Query Editor.

2. Identify and Remove Null Values

Steps:

- Check for **null values** in the "Sales" and "Quantity" columns.
- Remove or replace null values using appropriate Power Query functions.

Expected Output:

- Null values are handled using appropriate techniques (e.g., replacing with 0 or median value).

3. Remove Duplicates and Fix Errors

Steps:

- Identify and **remove duplicate** entries from the "Customer ID" column.
- Fix any **data entry errors** (e.g., incorrect category names).

Expected Output:

- Duplicate Customer IDs are removed.
- Errors in data entries (e.g., misspelled category names) are corrected.

4. Adjust Data Types for Analysis

Steps:

- Ensure "Date" is in **Date format**.

- Convert "Sales" and "Quantity" to **Whole Number or Decimal** format.
- Ensure "Order ID" and "Customer ID" are in **Text format**.

 Expected Output:

- Data types are correctly formatted for analysis.

5. Transform Data Using Power Query Functions

 Steps:

- Create a **calculated column** for **Total Revenue** ($\text{Sales} \times \text{Quantity}$).
- Extract **year from the Date column** to enable time-based analysis.
- Rename columns for **clarity and consistency**.

 Expected Output:

- A new **Total Revenue column** is created.
- A **Year column** is extracted from the Date.
- Columns have clear and meaningful names.